Special Task

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- Domain Name: AIML Development
- College Name: Zeal Polytechnic
- Branch Name: AIML

Experiment: Employee Dataset – Data Reading, Loading, Exploration and Understanding.

OUTPUT:

Step 1: Import Libraries & Upload Dataset. (Google Colab)

```
import numpy as np
import pandas as pd
```

Double-Click (Or enter) to edit

```
from google.colab import files
uploaded = files.upload()
```

```
Choose files Employee.csv
Employee.csv(text/csv) - 4238 bytes, last modified: 06/07/2026 - 100% done
Saving Employee.csv to Employee.csv
```

Step 2: Read Dataset

```
df = pd.read_csv("Employee.csv")
df
```

	Emp_ID	Name	Department	Age	Experience_Years	Salary_INR	City
0	1001	Employee_1	HR	23	1	306500	Mumbai
1	1002	Employee_2	Finance	24	2	313000	Nagpur
2	1003	Employee_3	Sales	25	3	319500	Nashik
3	1004	Employee_4	Marketing	26	4	326000	Kolhapur
4	1005	Employee_5	IT	27	5	332500	Thane
...	...	...	...	...	...	...	...
95	1096	Employee_96	HR	28	0	924000	Pune
96	1097	Employee_97	Finance	29	1	930500	Mumbai
97	1098	Employee_98	Sales	30	2	937000	Nagpur
98	1099	Employee_99	Marketing	22	3	943500	Nashik
99	1100	Employee_100	IT	23	4	950000	Kolhapur

100 rows × 7 columns

Step 3: Read Dataset & Display First 5 Rows

```
import pandas as pd

df = pd.read_csv("Employee.csv")
df.head()
```

	Emp_ID	Name	Department	Age	Experience_Years	Salary_INR	City
0	1001	Employee_1	HR	23	1	306500	Mumbai
1	1002	Employee_2	Finance	24	2	313000	Nagpur
2	1003	Employee_3	Sales	25	3	319500	Nashik
3	1004	Employee_4	Marketing	26	4	326000	Kolhapur
4	1005	Employee_5	IT	27	5	332500	Thane

Step 4: Display Last 5 Rows

```python
df = pd.read_csv("Employee.csv")
df.tail()
```

1 to 5 of 5 entries Filter

index	Emp_ID	Name	Department	Age	Experience_Years	Salary_INR	City
95	1096	Employee_96	HR	28	0	924000	Pune
96	1097	Employee_97	Finance	29	1	930500	Mumbai
97	1098	Employee_98	Sales	30	2	937000	Nagpur
98	1099	Employee_99	Marketing	22	3	943500	Nashik
99	1100	Employee_100	IT	23	4	950000	Kolhapur

Step 5: Shape of Dataset

```python
df = pd.read_csv("Employee.csv")
df.shape
```

```
(100, 7)
```

Step 6: Display Column Names

```python
df = pd.read_csv("Employee.csv")
df.columns
```

```
Index(['Emp_ID', 'Name', 'Department', 'Age', 'Experience_Years', 'Salary_INR',
       'City'],
      dtype='object')
```

Step 7: Dataset Information

```
df = pd.read_csv("Employee.csv")
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Emp_ID            100 non-null    int64
 1   Name              100 non-null    object
 2   Department        100 non-null    object
 3   Age               100 non-null    int64
 4   Experience_Years  100 non-null    int64
 5   Salary_INR        100 non-null    int64
 6   City              100 non-null    object
dtypes: int64(4), object(3)
memory usage: 5.6+ KB
```

Step 8 : Statistical Summary

```
df = pd.read_csv("Employee.csv")
df.describe()
```

	Emp_ID	Age	Experience_Years	Salary_INR
count	100.000000	100.000000	100.000000	100.000000
mean	1050.500000	25.970000	3.460000	628250.000000
std	29.011492	2.599359	2.276006	188574.697843
min	1001.000000	22.000000	0.000000	306500.000000
25%	1025.750000	24.000000	1.750000	467375.000000
50%	1050.500000	26.000000	3.000000	628250.000000
75%	1075.250000	28.000000	5.000000	789125.000000
max	1100.000000	30.000000	7.000000	950000.000000

Step 9: Data Types

```
df = pd.read_csv("Employee.csv")
df.dtypes
```

	0
Emp_ID	int64
Name	object
Department	object
Age	int64
Experience_Years	int64
Salary_INR	int64
City	object

dtype: object

Step 10: Missing Values

```
df = pd.read_csv("Employee.csv")
df.isnull().sum()
```

	0
Emp_ID	0
Name	0
Department	0
Age	0
Experience_Years	0
Salary_INR	0
City	0

dtype: int64

Step 11: Unique Values

```
df = pd.read_csv("Employee.csv")
df.nunique()
```

	0
Emp_ID	100
Name	100
Department	5
Age	9
Experience_Years	8
Salary_INR	100
City	6

dtype: int64

Step 12: Random Sample

```
df = pd.read_csv("Employee.csv")
df.sample(10)
```

	Emp_ID	Name	Department	Age	Experience_Years	Salary_INR	City
29	1030	Employee_30	IT	25	6	495000	Pune
91	1092	Employee_92	Finance	24	4	898000	Nagpur
62	1063	Employee_63	Sales	22	7	709500	Nashik
97	1098	Employee_98	Sales	30	2	937000	Nagpur
99	1100	Employee_100	IT	23	4	950000	Kolhapur
66	1067	Employee_67	Finance	26	3	735500	Mumbai
15	1016	Employee_16	HR	29	0	404000	Kolhapur
12	1013	Employee_13	Sales	26	5	384500	Mumbai
49	1050	Employee_50	IT	27	2	625000	Nagpur
76	1077	Employee_77	Finance	27	5	800500	Thane

Step 13: Value Counts

```
df = pd.read_csv("Employee.csv")
df.value_counts()
```

Emp_ID	Name	Department	Age	Experience_Years	Salary_INR	City	count
1001	Employee_1	HR	23	1	306500	Mumbai	1
1002	Employee_2	Finance	24	2	313000	Nagpur	1
1003	Employee_3	Sales	25	3	319500	Nashik	1
1004	Employee_4	Marketing	26	4	326000	Kolhapur	1
1005	Employee_5	IT	27	5	332500	Thane	1
...	...	...	...	...	...	...	...
1096	Employee_96	HR	28	0	924000	Pune	1
1097	Employee_97	Finance	29	1	930500	Mumbai	1
1098	Employee_98	Sales	30	2	937000	Nagpur	1
1099	Employee_99	Marketing	22	3	943500	Nashik	1
1100	Employee_100	IT	23	4	950000	Kolhapur	1

100 rows × 1 columns

dtype: int64